8^{th} Grade Unit 1 Assessment Guide

General Information	This assessment can be used by the teacher as a short diagnostic to determine whether a student recalls some of the prerequisite benchmarks from grade 7 or it can be used after a class completes unit 1. The teacher is encouraged to remember that the questions on this assessment cannot give a teacher a complete picture, but they can serve as general information. Just because a student misses a question on this assessment does not mean that the student has a learning gap. Teachers should look at their students as a collective to determine if there are any noticeable gaps that may take some remediation. Often remediation is best done within the grade-level learning as students continue to learn new content. Students may need support just prior to or during the grade-level learning. Halting all grade-level learning to reteach content from earlier grades may not be the best form of remediation. National Council of Teachers of Mathematics' Equity and Access statement includes the phrase "all students routinely have opportunities to experience high-quality mathematics instruction, learn challenging mathematics content, and receive the support necessary to be successful." This introductory unit and assessment can be used as a tool to help students receive support while continuing their learning of challenging mathematics content. If this assessment is used as a diagnostic, then the content from unit 1 may be a tool for the teacher to use to supplement student learning to help remediate. Teachers may also use instructional tasks from the grade 7 Big M for the prerequisite benchmarks.
Calculator Information	The questions on this assessment are no calculator. Teachers may decide to split the test into a calculator and no calculator portion, if desired. In this case, it is suggested that questions 1-11 remain no calculator, but that students have access to a scientific calculator for questions 12-17.
Total Recommended Points	The total recommended points in this guide is 99 points. Teachers can either determine where best to use the extra 1 point or round the total value up to 100 points.

<table>
<tr><th>Question</th><th>1</th><th>Benchmark(s)</th><th>MA.7.NSO.1.2</th></tr>
<tr><td colspan="2">

Fraction	Decimal	Percent
$\frac{9}{25}$	0.36	36%

</td><td>Recommended Scoring

(4 Total Points)</td><td>Consider giving students 2 points for each equivalent rational number in the correct form. There should be no expectation of “reducing to lowest terms” or simplifying the answer. Please note that students could write $\frac{9}{25}$ as $\frac{36}{100}$ or $\frac{18}{50}$.</td></tr>
</table>

<table>
<tr><th>Question</th><th>2</th><th>Benchmark(s)</th><td>MA.7.NSO.1.2</td></tr>
<tr><td colspan="2">
<table>
<tr><th>Fraction</th><th>Decimal</th><th>Percent</th></tr>
<tr><td>$\frac{58}{99}$</td><td>$0.\overline{58}$</td><td>$58.\overline{58}\%$</td></tr>
</table>
</td><th>Recommended Scoring

(4 Total Points)</th><td>Consider giving students 2 points for each equivalent rational number in the correct form. There should be no expectation of writing the repeating decimal to a particular place value, but 1 point should be deducted if the recurring digits are not correctly represented under a horizontal line. Please note that students could write $0.\overline{58}$ as $0.58\overline{58}$, $0.\overline{5858}$, or any other correct representation. The same applies to the percent form of the repeating decimal.</td></tr>
</table>

Question	3	Benchmark(s)	MA.7.NSO.1.2
<table><tr><th>Fraction</th><th>Decimal</th><th>Percent</th></tr><tr><td>$\frac{7}{200}$</td><td>0.035</td><td>3.5%</td></tr></table>		**Recommended Scoring** **(4 Total Points)**	Consider giving students 2 points for each equivalent rational number in the correct form.
Question	**4**	**Benchmark(s)**	MA.7.NSO.1.2
<table><tr><th>Fraction</th><th>Decimal</th><th>Percent</th></tr><tr><td>$\frac{7}{25}$</td><td>0.28</td><td>28%</td></tr></table>		**Recommended Scoring** **(4 Total Points)**	Consider giving students 2 points for each equivalent rational number in the correct form. There should be no expectation of "reducing to lowest terms" or simplifying the answer. Please note that students could write $\frac{7}{25}$ as $\frac{28}{100}$ or $\frac{14}{50}$.

Question	5	Benchmark(s)	MA.7.NSO.1.1
Select all of the equivalent expressions. ☐ 0 ☐ 8.1 ☐ $(2.4) \cdot 1000$ ☐ $(0.6)^4 \cdot (10)^3$ ☐ $(0.081) \cdot 1000$ ☐ $(10 \cdot 0.6)^3 \cdot 10$ ☐ $((0.6)^2)^2 \cdot (10)^3$ ☐ $\frac{(0.6)(0.6)(0.6)(0.6)}{0.2} \cdot (10)(10)(10)$		**Recommended Scoring** **(8 Total Points)**	Consider giving students 4 points for each correct answer and deduct a point for each incorrect answer. A student who selects all of the answers should not receive any credit.
Question	**6**	**Benchmark(s)**	MA.7.NSO.2.1
Answer	548	**Recommended Scoring** **(8 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**7**	**Benchmark(s)**	MA.7.NSO.2.1
Answer	$\frac{6400}{729}$	**Recommended Scoring** **(5 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	8	Benchmark(s)	MA.7.NSO.2.1
Answer -3.5		**Recommended Scoring** **(5 Total Points)**	Students may provide an equivalent answer. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**9**	**Benchmark(s)**	MA.7.NSO.2.1
Answer 121		**Recommended Scoring** **(8 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**10**	**Benchmark(s)**	MA.7.NSO.2.1
Answer -27		**Recommended Scoring** **(6 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	11	Benchmark(s)	MA.7.NSO.2.1
Answer: -0.5		Recommended Scoring (5 Total Points)	Students may provide an equivalent answer. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**12**	**Benchmark(s)**	MA.7.NSO.2.2
$w =$ -34		Recommended Scoring (5 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**13**	**Benchmark(s)**	MA.7.NSO.2.2
$n =$ 33.6		Recommended Scoring (5 Total Points)	Students may provide an equivalent answer. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	14	Benchmark(s)	MA.7.NSO.2.2
$k =$ $4\frac{11}{30}$		Recommended Scoring (5 Total Points)	Students may provide an equivalent answer. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	15	Benchmark(s)	MA.7.NSO.2.2
$x =$ -11.3		Recommended Scoring (5 Total Points)	Students may provide an equivalent answer. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.

Question	16a	Benchmark(s)	MA.7.NSO.2.2
Equation $12.65h + 246.75 = 550.35$		Recommended Scoring (5 Total Points)	No partial credit is suggested.

Question	16b	Benchmark(s)	MA.7.NSO.2.2
Number of Hours 24		**Recommended Scoring** (4 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**17a**	**Benchmark(s)**	MA.7.NSO.2.2
$\boxed{9}\,(m + \boxed{5.2}) = \boxed{148.5}$		**Recommended Scoring** (5 Total Points)	No partial credit is suggested
Question	**17b**	**Benchmark(s)**	MA.7.NSO.2.2
Number of Miles 11.3		**Recommended Scoring** (4 Total Points)	Students may provide an equivalent answer. Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.